

Response Letter with Annotations
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Kozlowski, A. C., Taddy, M., & Evans, J. A. (2019). The Geometry of Culture: Analyzing the Meanings of Class through Word Embeddings. American Sociological Review, 84(5), 905-949. <https://doi.org/10.1177/0003122419877135> (Original work published 2019)

Annotations

1. Purposes

a. Motivation (one-sentence summary)

The purpose of this paper was to present a case example of a new “emerging computational approach” known as **neural-network word embedding models**, where the authors apply this method to understand cultural dimensions of social class and how these definitions evolved throughout 20th century literature.

b. Specific Rationale (several bullet points)

- The authors wanted to apply computational methods to analyze social class from a sociological perspective while considering “cultural dimensions”, where words, objects, people, and social groups carry context and meaning tied to class structures.
- The authors were interested in using large, digitized text data to derive meaning and describe relationships between words, rather than utilizing already existing methods such as semantic network analysis and topic modeling which are not sufficient at deriving meaning/context.

c. Research Questions and/or Hypotheses

- How did collective understandings of social class evolve in the United States over the course of the 20th century?

2. Method

a. Research Design

- The neural-network embedding model uses a type of “vector space model” (Word2vec) to arrange words appearing in a corpus into one large matrix, where words are positioned near each other based on similar contexts.
- Distance between words is measured diagonally rather than straight line distance (Euclidean)
- The model derives dimensions in these vector spaces which apply to the cultural dimensions of interest to the study team, arranged on a spectrum of antonyms (i.e. affluence – poverty), then words from the corpus or projected onto the spectrum based on their position in context relative to each end of the antonym’s spectrum.
- Multiple dimensions can be used at once to place words on two antonym spectrums (i.e., affluence – poverty and feminine – masculine)
- The study team trains three different models, two using word2vec embeddings on Google Ngrams and Google News, and a third trained on data scraped from the Common Crawl with GloVe embeddings.

b. Sample

- The sample data used to train the model utilized “millions of books published over the entire twentieth century” digitized into the Google Ngram corpus – ultimately contains “text from 6 percent of all books ever published”
- The corpus was divided by decade from years 1900 to 1999, resulting in 10 word embedding models
- The model collected 5-grams (any sequence of five words that occurs more than 40 times over the entirety of scanned texts) to obtain enough context to derive meaning

c. Measures

- Model dimensions were sorted by the model into five cultural dimensions of class informed by sociological theory: Affluence, Employment, Status, Education, Cultivation, and two additional cultural dimensions: Morality, and Gender.
- Cultural dimensions were assessed by fielding a survey (N = 398) where respondents assessed various items in topical domains on a spectrum of “very working class” to “very upper class” on Mechanical Turk, resulting in ratings of different items on a 100 point scale.
- Antonym pairs were assembled by referencing five thesauri: three contemporary, and two historical from the early 1900s.

d. Analysis

- Pearson’s correlation coefficients were calculated to compare and validate the survey estimates and word embedding estimates between the three different word embedding models, where the Google Ngrams corpus performed poorly, especially on racial associations – and was excluded from further analyses
- Plots corresponding to each embedding projection were generated in order to display Pearson’s correlations for the models, and cosine similarity for cultural dimension measures. OLS regression models were used to predict affluence projections from other cultural dimensions.
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3. Results and Conclusions

a. Answers to RQs and Hypotheses

- The word embedding models performed well with statistically significant, positive associations between the model and survey responses.
- The cultural dimensions most parallel to affluence were Cultivation and Status, followed by morality, gender, and education, and these relationships were fairly stable over the study period.
- There was a weakly negative association between the education dimension and affluence from 1900-1960s which the authors believed education’s association with affluence was more so a “byproduct of its association with cultivation, sophistication, and refinement”, as well as the twentieth century context that education was sometimes implied as a necessity to enter the workforce rather than a privilege.

- Education was heavily associated with affluence by 1990 and much more closely associated with the cultivation dimension.
- Relationships between cultural dimensions remained fairly constant throughout the 20th century – where cultivation, morality, and education are closely related, and orthogonal to employment.
- Employment has a negative relationship with morality, but is positively associated with the affluence and status dimensions.
- Within dimension relationships between decades revealed that decades are closely correlated, but comparisons between 1900 and 1990 correlation weakens by .2-.3 – the authors suggest that this pattern shows how stable class dimensions are over time, but also demonstrates the nuance and evolution of how culture evolves over time.
- The authors look more closely at employment due to its declining correlation with other dimensions throughout the 1900s, which revealed the difference in employment titles throughout the century – demonstrating the impact of industrialization on cultural language (i.e. lords and earls were replaced by moguls and promoters)
- Language such as wage, and earners are common in the beginning of the century and fades as the century goes on as the workforce switches to salary and contract positions.

b. Include Key Figures/Tables

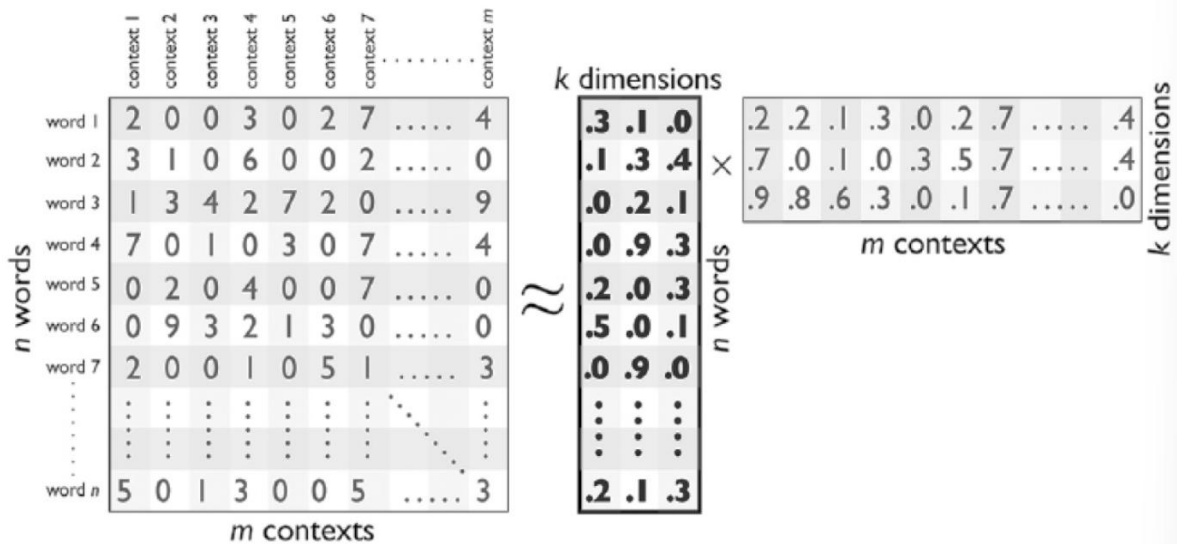


Figure 1. Schematic Illustration of the Descriptive Problem Neural Word Embeddings Solve—How to Represent All Words from a Corpus within a k -Dimensional Space That Best Preserves Distances between Words in Their Local Contexts

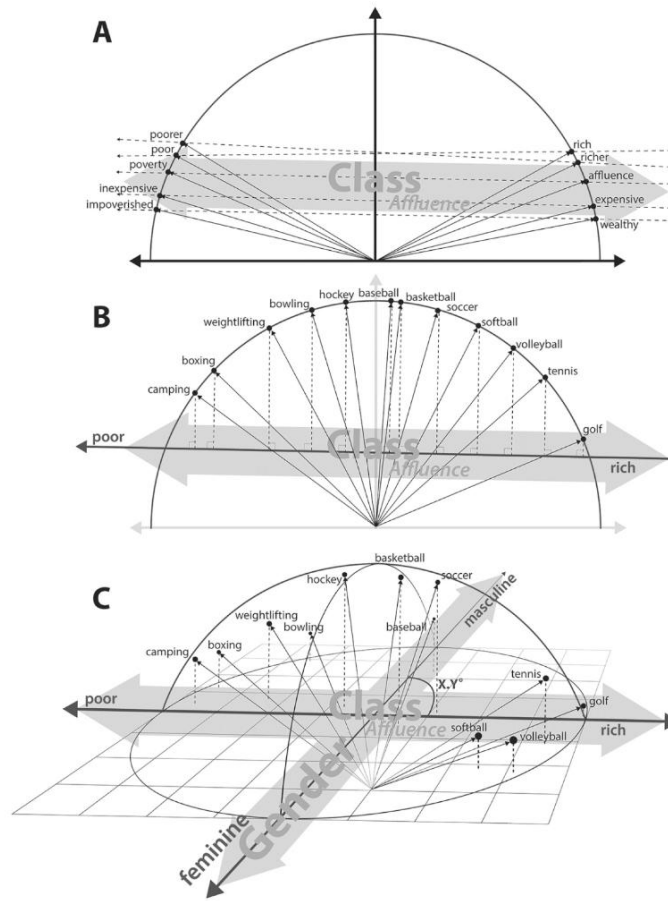


Figure 2. Conceptual Diagram of (A) the Construction of a Cultural Dimension; (B) the Projection of Words onto That Dimension; and (C) the Simultaneous Projection of Words onto Multiple Dimensions

Table 1. Pearson Correlations between Survey Estimates and Word Embedding Estimates for Gender, Class, and Race Associations

	Class (Affluence)	Gender	Race
Google Ngrams <i>word2vec</i> Embedding [†]	.53	.76	.27
Google News <i>word2vec</i> Embedding	.58	.88	.75
Common Crawl <i>GloVe</i> Embedding	.57	.90	.44

Note: $N = 59$, except [†] $N = 58$ where one word measured in the survey did not occur frequently enough in the text to appear in the word embedding.

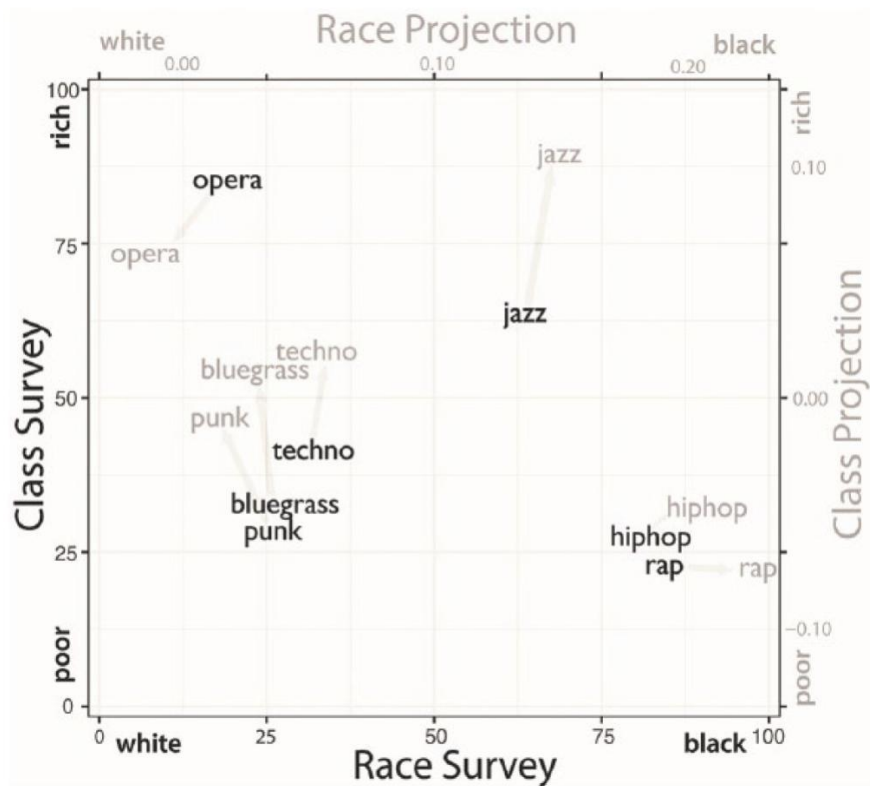


Figure 3. Projection of Music Genres onto Race and Class Dimensions of the Google News Word Embedding (Gray) and Average Survey Ratings for Race and Class Associations (Black)

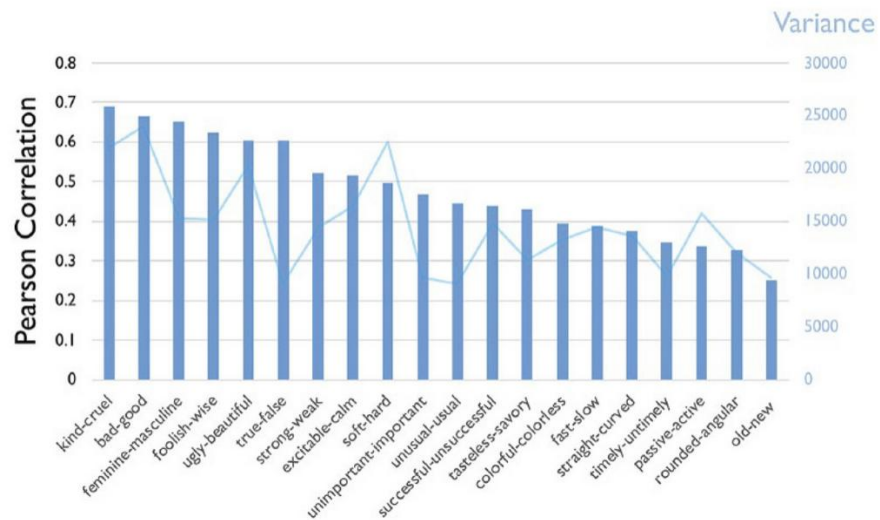


Figure 4. Correlations between Word Embedding Projections and Human-Rated Associations on 20 Semantic Dimensions, Alongside Variance of Average Human-Ratings on Those Dimensions; 1950 to 1959 Google Ngrams Corpus

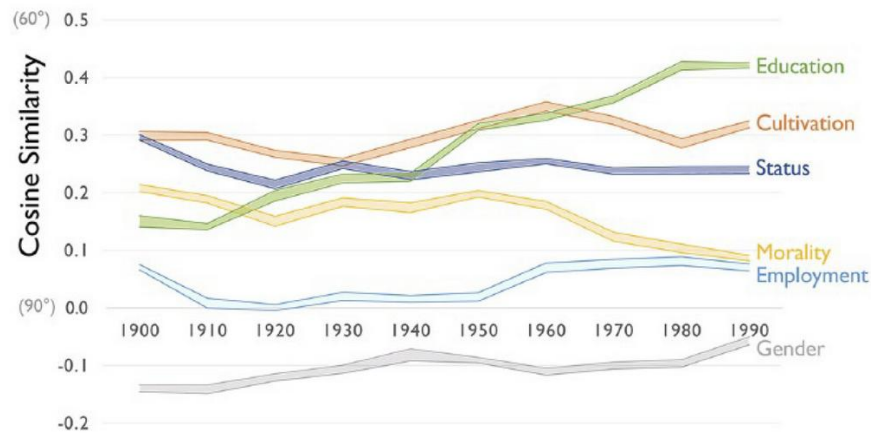


Figure 5. Cosine Similarity between the Affluence Dimension and Six Other Cultural Dimensions of Class by Decade; 1900 to 1999 Google Ngrams Corpus
Note: Bands represent 90 percent bootstrapped confidence intervals produced by subsampling.

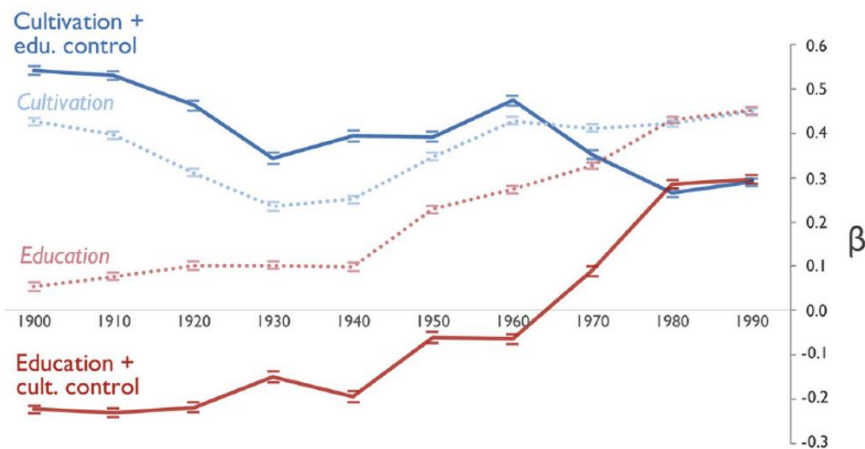


Figure 6. Standardized Coefficients from OLS Regression Models in Which Word Projections on Cultivation and Education Dimensions Predict Projection on the Affluence Dimension; 1900 to 1999 Google Ngrams Corpus
Note: A separate OLS regression model is fit for each decade; $N = 50,000$ most common words in each decade.

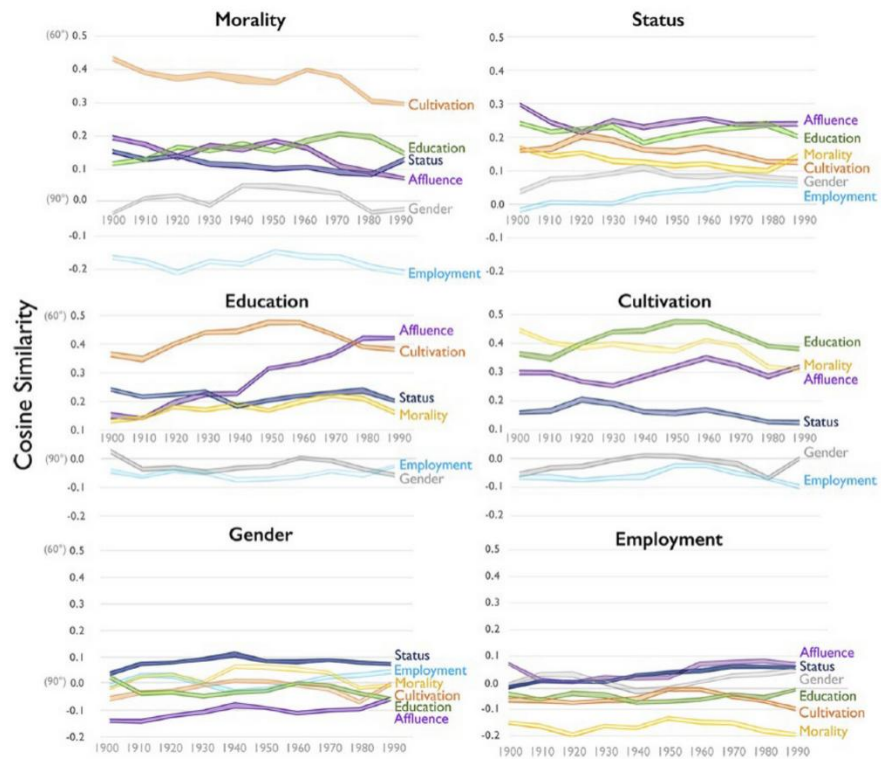


Figure 7. Cosine Similarity between Each Class Dimension and All Others by Decade; 1900 to 1999 Google Ngrams Corpus
Note: Bands represent 90 percent confidence intervals produced by subsampling.

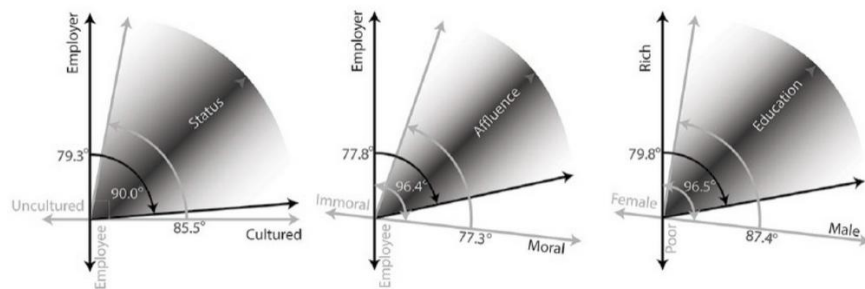


Figure 8. Conceptual Diagram of the Distortions Introduced When Reducing High-Dimensional Embeddings to Two-Dimensions

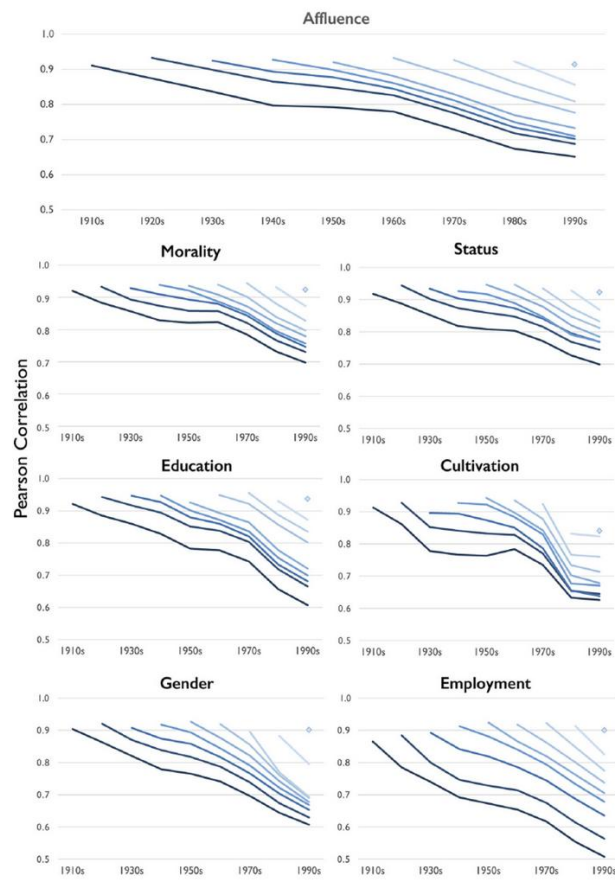


Figure 9. Correlation of 50,000 Most Common Words' Projection in One Decade with Their Projection in Each Subsequent Decade for Seven Cultural Dimensions of Class; 1900 to 1999 Google Ngrams Corpus

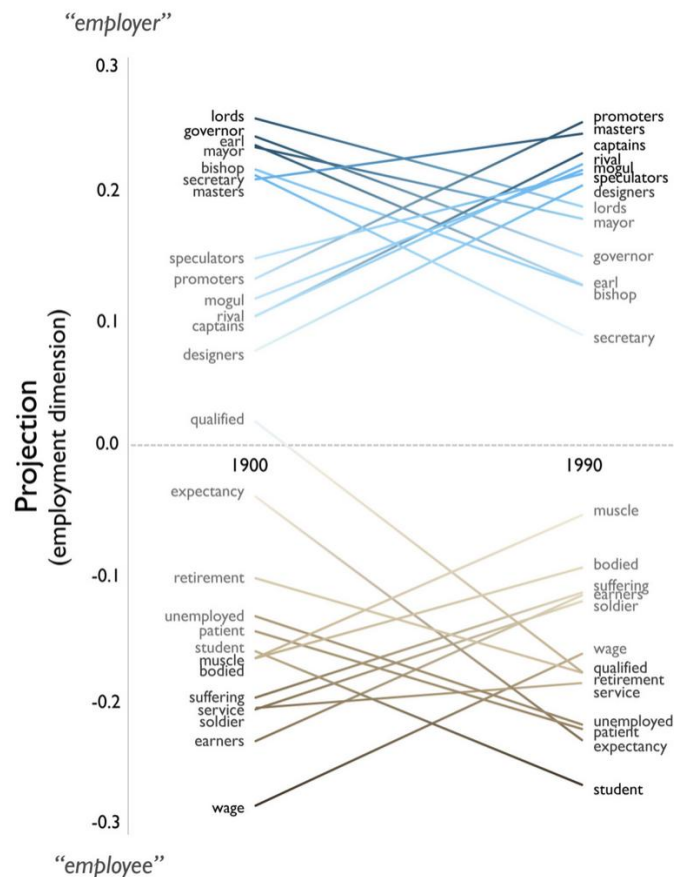


Figure 10. Words That Project High and Low on the Employment Dimension of Word Embedding Models Trained on Texts Published at the Beginning and End of the Twentieth Century; 1900–1919 and 1980–1999 Google Ngrams Corpus

c. Authors' Conclusions

The authors concluded that their word embeddings model worked well in conducting a macro-level analysis of cultural dimensions while preserving the context and complexity of how these words were used throughout the 20th century. They found that the cultivation, morality, and education dimensions were closely related and were negatively associated with classic “Marxian” owner/worker relationships.

4. Discussion

a. Implications

- The authors state that this study was only one example of possible use cases for word embedding models in analyzing large text corpora – which could range far beyond historical literature in topic and source material.
- Comparing cultural dimensions is also a very narrow application of using vector spaces and assessing similarity.

b. Limitations

- Word embeddings require a very large corpus in order for the vector space to capture the level of subtlety and complexity of associations between words – requiring several million words or more for the math to work out.
- The algorithms that underlie the training of word embedding models are very complex and abstract, which, in turn, makes the interpretation of these models immensely abstract. These results are difficult to interpret and convey findings to those unfamiliar with these processes – and also create issues with diagnosing abnormalities/distortions if the researcher's understanding of the model under the hood is lacking.
- The books digitized by Google are not a random or representative sample of U.S. culture broadly – where those who were able to publish in the 1900s were from more elitist and predominantly white backgrounds.
- Word embeddings cannot identify cultural dimensions that go beyond the individual word association level – aka, they cannot assess system or societal level context.

c. Future Directions

- Word embedding models have limitations, but when applied appropriately to very large corpora of text, can begin to parse apart meaning and association between words, and even assess changes over time. These models can provide detailed insight for larger corpora than topic modeling or other textual analysis methods can handle reasonably as well.

Response Letter (~600 words)

- Your assessment of the study

The Kozlowski et al. (2019) study was a very interesting and novel approach to managing a massive dataset and deriving meaning beyond what a topic model or data dictionary could do, which is no easy task. The authors clearly spent a great deal of time and effort in figuring out ways to validate this method to the best of their ability, and their approach does seem to fill a gap in the wide array of textual analysis methods that already exist. As a population studies scholar, I was thinking about all the ways the historians and demographers I talk to would be interested in the longitudinal aspect of this study – how cool would it be to look at narrative autopsy transcripts or various census records in some variation of this method? I am of course drawn to news data applications of this method, though I have my reservations about the method as follows:

- Weaknesses

My main concern with this word embedding model method is the inherent limitation of binary characterizations, even on a spectrum as the model demonstrates in Figure 2. Computational methods of course rely on an ability to turn subjective texts into some form of definitive objects parse-able by a machine, which is already very difficult to do. But I fear what conclusions researchers may draw from defining these “cultural dimensions” as binary spectrums, especially when Kozlowski et al. derive their idea of these dimensions in part from “unconscious biases”. I think that their cultural dimensions measure heuristic associations – the snap judgments that humans make upon first glance of things without thinking more deeply about context – rather than capturing true cultural context and meaning. The meaning came from the study team’s interpretations, as research findings almost always do.

My other major concern with this work is its interpretability – I can somewhat understand how this model operates conceptually, but cosine similarity is a difficult measure to explain to others without the full preface of how this model works. I think the figures and conceptual models do help with explaining how the model is meant to be assessed. This definitely isn’t a model to use if you’re wanting to convey information about a large corpus to a lay audience.

I have other minor concerns which the authors did address as limitations of the work, such as the inherent bias in 20th century literature as to who was able to publish and write. I appreciated their acknowledgement of the fact that models can’t account for broader sociological context (as someone from a medical sociology background) as that requires human expertise. Overall, I would say this model is not easy to work with, but given the right kind of training data and a clear understanding of what one wants to measure, this type of modeling can provide novel insights and open up an area of computational research.